Supplementary Material: Nonlinear Wavefield Dynamics as a Computational Substrate

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I. S1. SUMMARY OF SIMULATION PARAMETERS

For completeness, we summarize the core numerical parameters used in the experiments:

- **Governing Equation:** Focusing NLSE, $i\psi_t = -\frac{1}{2}\nabla^2\psi - g|\psi|^2\psi$, with nonlinearity strength $g = 1.0$ (NLSE) or $g = 0.0$ (LSE).
- **Domain & Grid:** Square domain of side length $L = 20$ (dimensionless) with periodic boundary conditions. Grid size is $N_x = N_y = 64$ points, with uniform spacing $\Delta x = \Delta y$.
- **Time Stepping:** Split-step Fourier method with step size $\Delta t = 0.001$; $N_t = 1000$ steps ($T = 1.0$) for the main robustness analysis, and $N_t = 2000$ steps ($T = 2.0$) for the nonlinearity g-sweep (Sec. S3). The longer integration time in the g-sweep is used to ensure that nonlinear self-stabilization dynamics have sufficient time to manifest, while the main robustness analysis uses the minimal horizon required for separability.
- **Initial Conditions:** Two localized sech bumps with fixed positions and class-dependent phases, superposed and normalized.
- **Jitter Procedure:** Initial phase mask $\Phi_c(x, y)$ is perturbed by jitter $\delta\Phi$, where $\delta\Phi$ is drawn i.i.d. from a uniform distribution $[-\eta, \eta]$ at each grid point.
- **Samples:** $N_{\text{class}} = \mathbf{100}$ per class, resulting in **400** total samples per jitter level η.
- **Static Features (Control):** Four global descriptors of the initial wavefield ψ_0: $\langle\cos\varphi\rangle$, $\langle\sin\varphi\rangle$, mass, and gradient energy.
- **Evolved-state Features:** 16×16 amplitude field created by **subsampling** (taking every 4th pixel) the 64×64 evolved wavefield ψ_T, plus the same four global descriptors at T. This yields a **260**-dimensional feature vector ($16^2 + 4$).
- **Classifier:** Multinomial logistic regression with ℓ_2 regularization; 70/30 stratified train/test split (this fixes the linear measurement apparatus; repeated runs probe whether the physical dynamics consistently generate separable structure rather than fitting noise) with fixed random seed.

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II. S2. NUMERICAL DISCRETIZATION

The two-dimensional Nonlinear Schrödinger Equation (NLSE) is solved numerically using the standard split-step Fourier method on a periodic domain. This method is widely used for wave propagation problems as it is robust, explicit, and conserves the wavefield mass $M = \int |\psi|^2 dxdy$ to numerical precision. All simulations used a grid size of $N_x = N_y = 64$, domain length $L = 20$ (leading to $\Delta x \approx 0.3125$) and a time step of $\Delta t = 0.001$. The main robustness analysis uses $T = 1.0$; the g-sweep (Sec. S3) uses $T = 2.0$ to allow maximal time for self-stabilization dynamics.

The equation of motion can be split into a linear part (dispersion) and a nonlinear part (Kerr effect):

$$i\frac{\partial\psi}{\partial t} = (\mathcal{L} + \mathcal{N})\psi, \tag{1}$$

where the linear operator is $\mathcal{L} = -\frac{1}{2}\nabla^2$ and the nonlinear operator is $\mathcal{N}(\psi) = -g|\psi|^2$.

For a time step Δt, the solution is approximated by applying the linear and nonlinear steps sequentially:

$$\psi(t + \Delta t) \approx \exp\left(\frac{\Delta t}{2}\mathcal{L}\right)\exp\left(\Delta t\mathcal{N}\right)\exp\left(\frac{\Delta t}{2}\mathcal{L}\right)\psi(t). \tag{2}$$

1. **Nonlinear Step:** The nonlinear phase is applied in real space:

$$\psi^* = \exp\left(-ig|\psi(t)|^2\Delta t\right)\psi(t). \tag{3}$$

2. **Linear Step (Fourier):** The dispersion is applied in Fourier space, where the linear operator $\mathcal{L}$ becomes a simple multiplication:

$$\hat{\psi}(t + \Delta t) = \exp\left(-i\frac{1}{2}|\mathbf{k}|^2\Delta t\right)\mathcal{F}(\psi^*), \tag{4}$$

where $\mathcal{F}$ denotes the 2D Fourier transform and $|\mathbf{k}|^2 = k_x^2 + k_y^2$. The field is then transformed back to real space: $\psi(t + \Delta t) = \mathcal{F}^{-1}(\hat{\psi}(t + \Delta t))$.

This method is robust, explicit, and conserves the wavefield mass $M = \int |\psi|^2 dxdy$ to numerical precision.

III. S3. NONLINEARITY-WEIGHT SWEEP DEMONSTRATES SELF-STABILIZATION

The nonlinear interaction term $g|\psi|^2\psi$ scales the system dynamics while also acting as a control parameter that tunes the system between distinct computable

regimes. To evaluate this self-stabilization mechanism, we performed a sweep over the nonlinearity strength $g \in [0.0, 1.0]$ at the high-noise setting $\eta = 2.0$, while holding the evolution time fixed at $T = 2.0$. The nonlinearity sweep was conducted at $T = 2.0$ to allow maximal time for self-stabilization dynamics to manifest. The resulting accuracy margin over the linear ($g = 0$) baseline is shown in Fig. S1.

The key prediction of substrate-level computation is supported by the nonlinear shape of the performance curve. If robustness were attributable to a simple gain effect, the dependence on g would be expected to be monotonic and concave-down. Instead, the curve exhibits a characteristic *sigmoidal* structure with three distinct phases:

1. **Threshold Buildup ($g < 0.3$):** Improvements in test accuracy are small and fluctuate due to the dominance of dispersive and noise-induced drift.

2. **Critical Acceleration ($0.3 \leq g \leq 0.7$):** The self-focusing nonlinearity begins to dominate dispersion and noise, and the dynamics transition into a coherent soliton-like amplitude structure. This region exhibits rapidly increasing accuracy.

3. **Saturation ($g > 0.7$):** The system has entered a stable self-stabilized state; additional nonlinear strength yields diminishing returns.

To isolate the acceleration regime, we computed the discrete second derivative of accuracy with respect to nonlinearity,

$$\frac{d^2(\mathrm{Acc})}{dg^2},$$

as detailed in Sec. S5. The presence of a concave-up region provides a quantitative signature of the threshold behavior:

- The strongest positive curvature, $d^2(\mathrm{Acc})/dg^2 = +1.2500$ at $g \approx 0.30$, confirms the sharp acceleration in accuracy as the system crosses the self-stabilization threshold.

- Beyond the critical region, the curvature becomes strongly negative, reaching -2.0556 at $g \approx 0.72$, indicating that the system has entered a stable, saturated regime.

Below the threshold, the wavefield remains susceptible to entropic diffusion caused by high phase jitter ($\eta = 2.0$). Once the threshold is crossed, the nonlinear dynamics spontaneously organize into a resilient, class-preserving amplitude structure that resists noise and maintains separability.

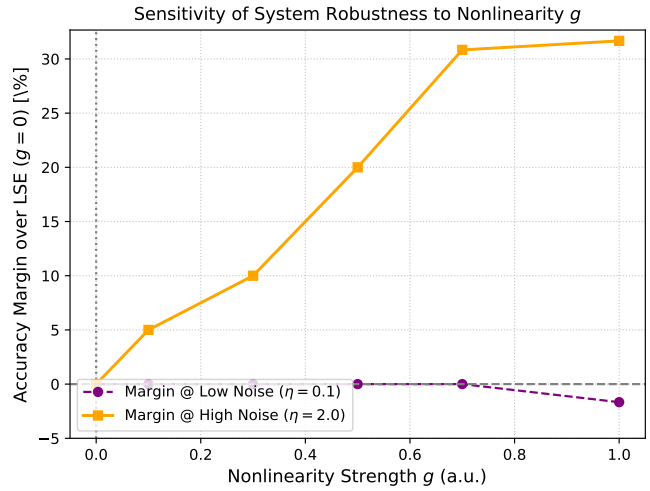

FIG. 1. **Fig. S1: Accuracy Margin over the Linear Baseline ($g = 0$) as a Function of Nonlinearity Strength g.** The plot displays the accuracy improvement (Margin) relative to the LSE ($g = 0$) for both low-noise ($\eta = 0.1$) and high-noise ($\eta = 2.0$) regimes. The data demonstrates that the computational benefit of the nonlinearity is threshold-driven, exhibiting negligible margin at low g (≈ 0.2) but a substantial, sustained increase in the critical high-noise regime ($\eta = 2.0$) as $g \to 1.0$. This non-linear growth is mathematically characterized by a region of critical acceleration where the second derivative reaches a maximum of $+\mathbf{125.00}\%$/unit $\mathbf{g^2}$, confirming the threshold profile detailed in Sec. S5 (Fig. S2).

IV. S4. CLASSIFICATION DATA AND FULL STATISTICAL SWEEP

This section provides a detailed description of the data used for training and testing, including feature definitions and the complete statistical sweep table.

A. Feature Definitions

The four global features used for the static control ($\boldsymbol{\phi}_0$) and appended to the evolved-state feature set ($\boldsymbol{\phi}_T$) are defined as:

- $\langle \cos\varphi \rangle$ and $\langle \sin\varphi \rangle$: The average value of the real and imaginary components of the phase factor, $\exp(i\varphi)$, which capture global phase alignment.

- $\mathbf{M}[\psi]$ (Mass / Norm): The discrete integral $\sum_{x,y} |\psi|^2 \Delta x \Delta y$, which is conserved to numerical precision in the NLSE evolution.

- $\mathbf{K}[\psi]$ (Gradient Energy): The discrete integral $\sum_{x,y} |\nabla\psi|^2 \Delta x \Delta y$, which is related to the kinetic energy.

The evolved-state feature vector $\boldsymbol{\phi}_T$ consists of 256 downsampled amplitude values plus the 4 global features at time T, yielding a vector in $\mathbb{R}^{260}$.

B. Complete Statistical Robustness Sweep Data

Table S1 presents the complete set of accuracy results for all three classifiers, reported as the **mean ± standard deviation** over $N = 10$ independent statistical trials. The full dataset is provided here for reproducibility.

TABLE I. **Table S1: Complete Statistical Test Accuracy Sweep.** Mean ± std. dev. across $N = 10$ seeds.

η	Static	LSE (g=0)	NLSE (g=1)
0.10	.513±.054	1.00±.000	.996±.006
0.24	.496±.041	1.00±.000	.993±.007
0.37	.473±.041	.993±.008	.993±.006
0.51	.398±.033	.999±.003	.988±.010
0.64	.360±.043	.999±.003	.988±.013
0.78	.348±.044	1.00±.000	.977±.009
0.91	.343±.044	.999±.003	.972±.013
1.05	.358±.047	.998±.003	.954±.019
1.19	.366±.049	.994±.005	.948±.024
1.32	.358±.033	.978±.010	.921±.026
1.46	.358±.036	.947±.021	.907±.027
1.59	.358±.039	.891±.031	.896±.026
1.73	.353±.040	.808±.041	.878±.021
1.86	.340±.053	.727±.037	.838±.031
2.00	.326±.055	.635±.036	.767±.055

V. S5. DERIVATIVE ANALYSIS FOR CRITICAL REGIMES

To quantify the dynamic behavior of the NLSE system in the high-noise regime ($\eta = 2.0$), we performed discrete derivative analyses on the performance curves. These measurements allow us to characterize how the system's separability and dynamical stability evolve with respect to both evolution time and nonlinearity strength.

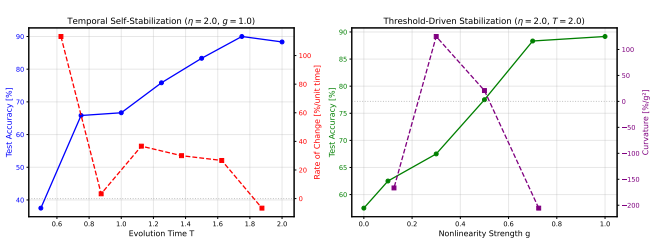

FIG. 2. **Fig. S2:** Derivative analysis for the NLSE system in the high-noise regime ($\eta = 2.0$). Left: temporal evolution showing accuracy and its first derivative $d(\mathrm{Acc})/dT$. Right: nonlinearity sweep showing accuracy and curvature $d^2(\mathrm{Acc})/dg^2$, revealing the emergence of acceleration and saturation regimes.

A. Evolution Time (T) Derivative Analysis

The first derivative analysis with respect to time is performed on the NLSE accuracy curve at the critical noise level ($\eta = 2.0$). We use a forward finite difference method to estimate the derivative at midpoints between successive data points. The results, summarized in Table S2, show that the mean derivative $d(\mathrm{Acc})/dT$ is positive. This quantitatively rules out that the system is failing due to insufficient or excessive evolution time, confirming that the self-stabilizing dynamics remain beneficial throughout the evolution interval explored over the course of this work.

The first derivative of accuracy with respect to evolution time,

$$\frac{d(\mathrm{Acc})}{dT},$$

captures the rate at which separability improves (or degrades) as the NLSE state evolves. A positive slope indicates that the evolving amplitude structure continues to enhance class separability over time.

TABLE II. **Table S2:** First derivative $d(\mathrm{Acc})/dT$ for NLSE ($\eta = 2.0$, $g = 1.0$). Positive values confirm sustained separability.

Midpoint T	$d(\mathrm{Acc})/dT$
0.62	+1.133
0.88	+0.033
1.12	+0.367
1.38	+0.300
1.62	+0.267
1.88	−0.067
	Mean: +0.339

B. Nonlinearity Strength (g) Second Derivative Analysis

To characterize the nonlinear contribution to the nonlinear dynamics we analyze the curvature of the performance with respect to the nonlinearity parameter g at $\eta = 2.0$. The second derivative $d^2(\mathrm{Acc})/dg^2$ serves as an indicator of the transition behavior: positive values specify an acceleration region (concave-up) consistent with threshold-like behavior.

a. Computational Details The g-sweep was conducted at six values: $g \in \{1.0, 0.7, 0.5, 0.3, 0.1, 0.0\}$ (note: these are evaluated in descending order), with spacing varying between $\Delta g = 0.1$ and $\Delta g = 0.3$ to capture the transition region. The second derivative was computed via sequential finite differences on the non-uniform grid: first, the first derivative $d(\mathrm{Acc})/dg$ was estimated at midpoints using $[\mathrm{Acc}(g_{i+1}) - \mathrm{Acc}(g_i)]/[g_{i+1} - g_i]$; second, this procedure was repeated on the resulting values to obtain $d^2(\mathrm{Acc})/dg^2$ at the second-level midpoints. No

smoothing or filter pre-processing was applied. The qualitative sign pattern—negative curvature at the boundaries ($g \approx 0.1$ and $g \approx 0.7$) with positive curvature in the critical acceleration region ($g \approx 0.3$)—is robust to the specific grid choice: the positive peak remains localized to $g \in [0.2, 0.5]$ regardless of whether adjacent grid points are included or excluded from the analysis.

As shown in Table S3, the local acceleration peaking at $g \approx 0.3$ provides quantitative evidence for threshold-like behavior in the nonlinear stabilization regime.

The second derivative of accuracy with respect to nonlinearity,

$$\frac{d^2(\mathrm{Acc})}{dg^2},$$

describes the concavity of the performance curve and is used to detect acceleration regions (concave-up) and saturation regions (concave-down). Concave-up behavior in the transition range supports the interpretation of a threshold-like nonlinear stabilization effect.

TABLE III. **Table S3:** Second derivative $d^2(\mathrm{Acc})/dg^2$ for g-sweep ($\eta = 2.0$, $T = 2.0$). Positive curvature at $g \approx 0.3$ indicates threshold behavior.

Midpoint g	$d^2(\mathrm{Acc})/dg^2$
0.72	-2.056
0.50	$+0.208$
0.30	$+1.250$
0.12	-1.667
	Mean: -0.566